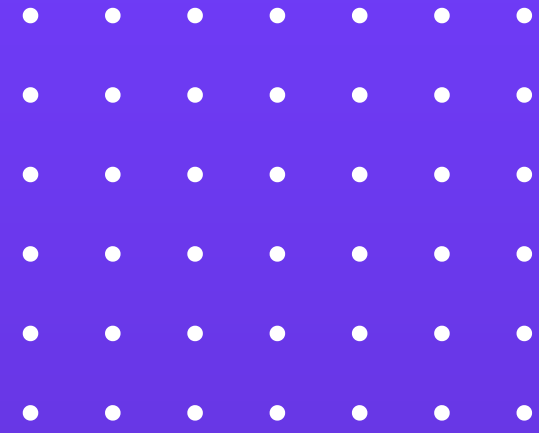


IoT Based Water Quality Control System



By

Student's Name

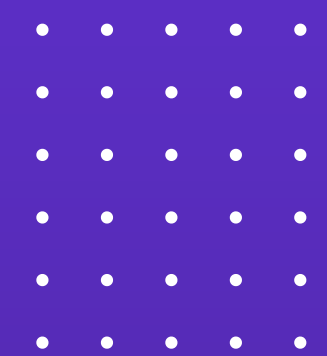
University Roll No

Under the guidance of

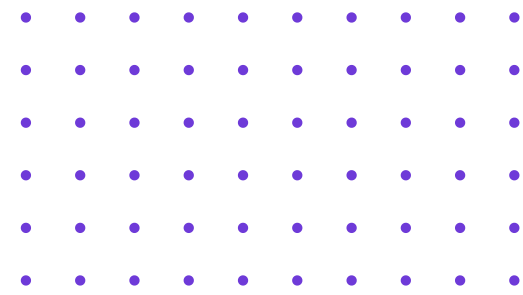
Debasis Dutta
Neha Bar
Asif khan
Liza Roy
Kiron Dey

16901619006
16901619047
16901619061
16901619005
16901619051

Assistant Professor
Sandipan Misra



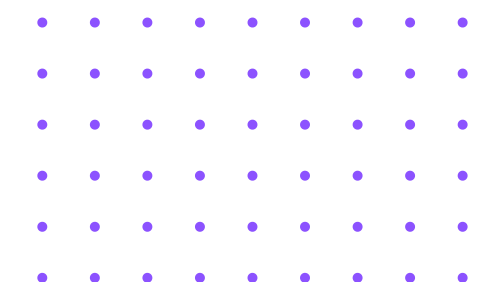
Objective



- The objective of this water quality monitoring system using internet of things is to find the quality of the water.
- To measure the water quality parameter such as pH, temperature, turbidity and TDS.
- Design a low-cost system for real-time monitoring of the water quality in IoT.
- System must be efficient as well as processing, sending and viewing data on cloud through WiFi to mobile.

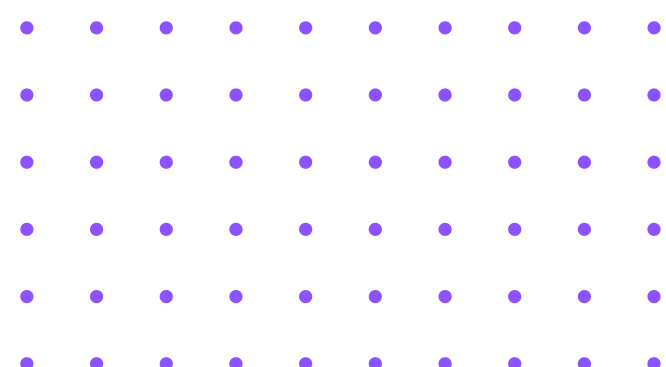


Abstract



Vision

Build a low-cost system for real-time monitoring of the water quality management in IOT.



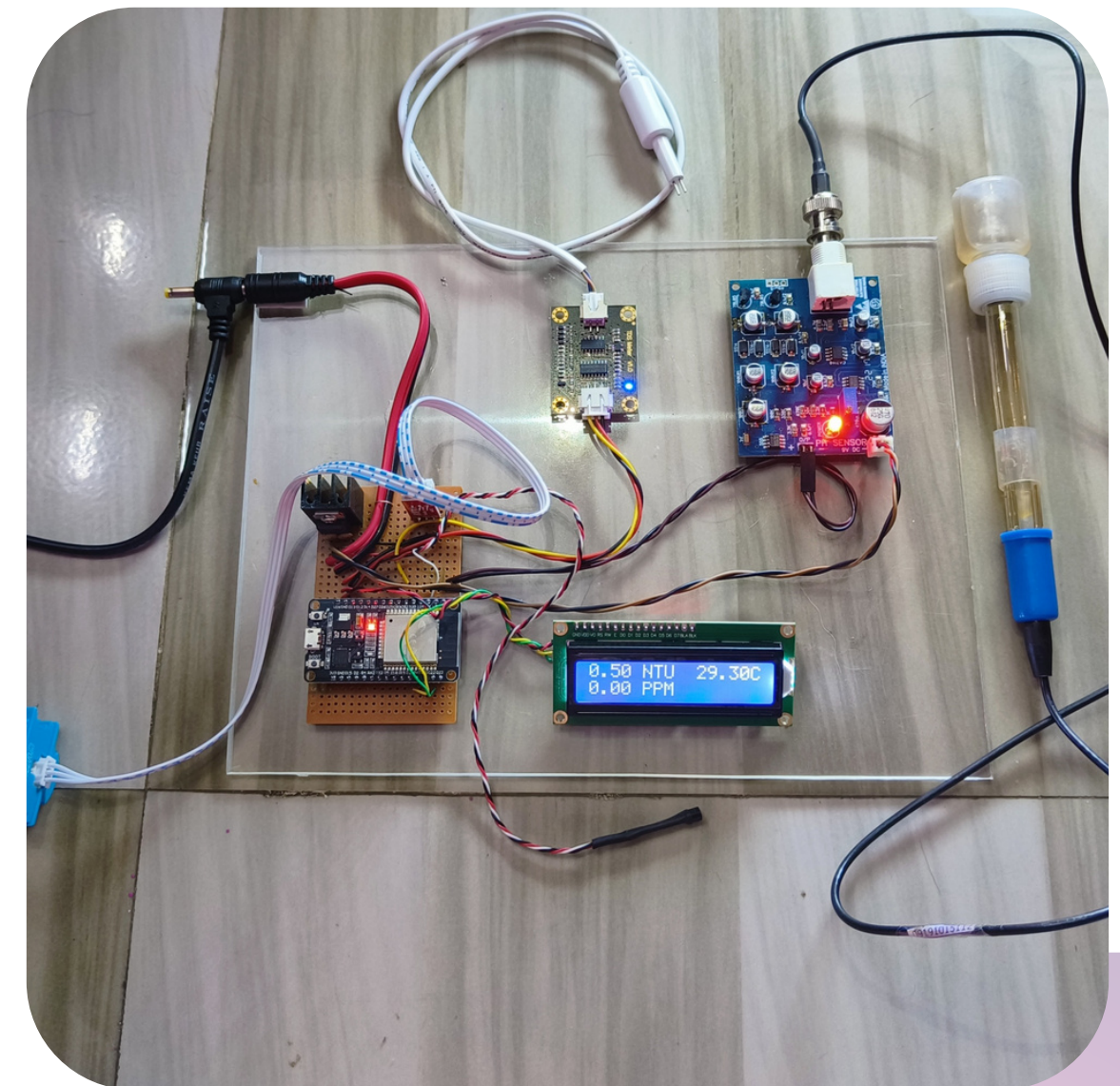
Mission

In order to guarantee the safe supply of the drinking water the quality needs to be monitor in real time.

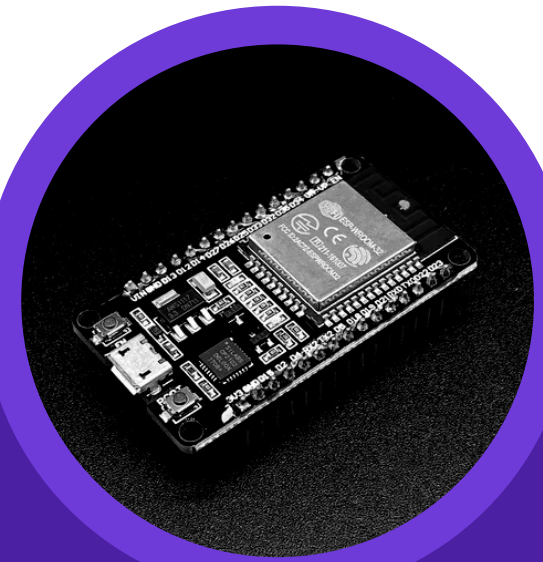


Project Design

In this, we present the theory on real time monitoring of water quality and quantity using IoT. The system consists of Esp32 microcontroller, different type of sensors like pH, TDS, turbidity sensor and Temperature sensor. The ESP32 is the main processor of the system which control and process the data generated by the sensors.



Components



ESP32

ESP32 is a low-cost, low-power Microcontroller with an integrated Wi-Fi and Bluetooth. It is the successor to the ESP8266 which is also a low-cost Wi-Fi microchip albeit with limited vastly limited functionality.



TDS SENSOR

TDS sensor/Meter Kit for measuring TDS value of the water. This product supports 3.3 ~ 5.5V wide voltage input, and 0 ~ 2.3V analog voltage output, which makes it compatible with 5V or 3.3V control systems or boards.



PH SENSOR

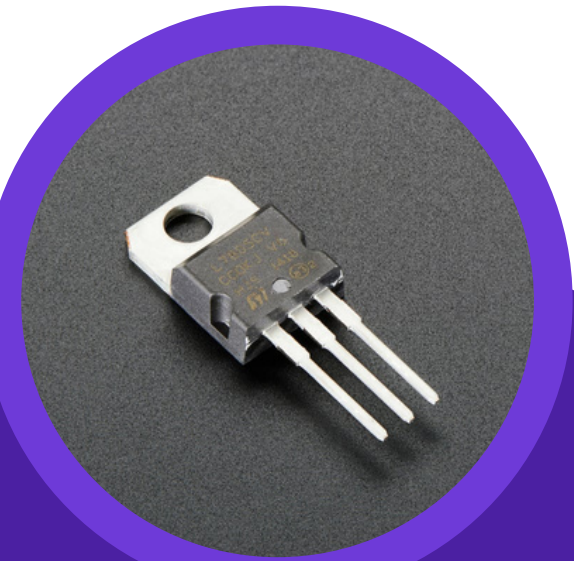
The pH meter might work as a voltmeter which measures the electric potential induced by the acidic solution. The pH difference is measured by comparing the produced electric potential with the known value and it is deducted.



TURBIDITY SENSOR

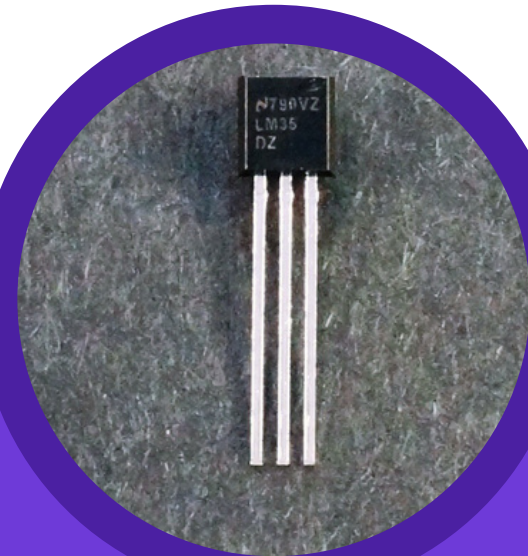
Turbidity sensors measure the amount of light that is scattered by the suspended solids in water. As the amount of total suspended solids (TSS) in water increases, the water's turbidity level (and cloudiness or haziness) increases.

Components



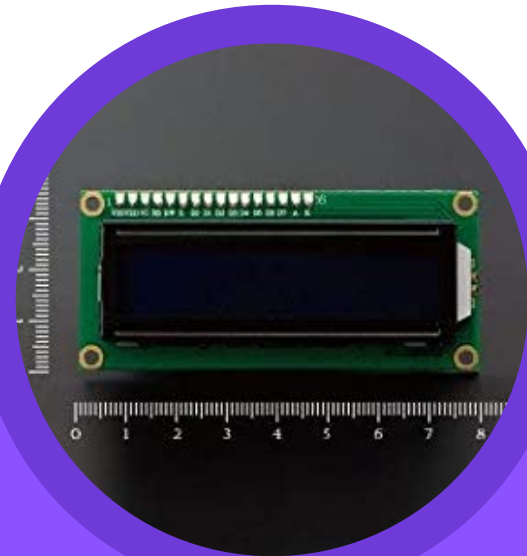
LM7805 REGULATOR

7805 IC, a member of 78xx series of fixed linear voltage regulators used to maintain such fluctuations, is a popular voltage regulator integrated circuit (IC). The xx in 78xx indicates the output voltage it provides.



LM35

It provides output voltage in Centigrade (Celsius). It does not require any external calibration circuitry. The sensitivity of LM35 is 10 mV/degree Celsius. As temperature increases, output voltage also increases.



I2C LCD DISPLAY

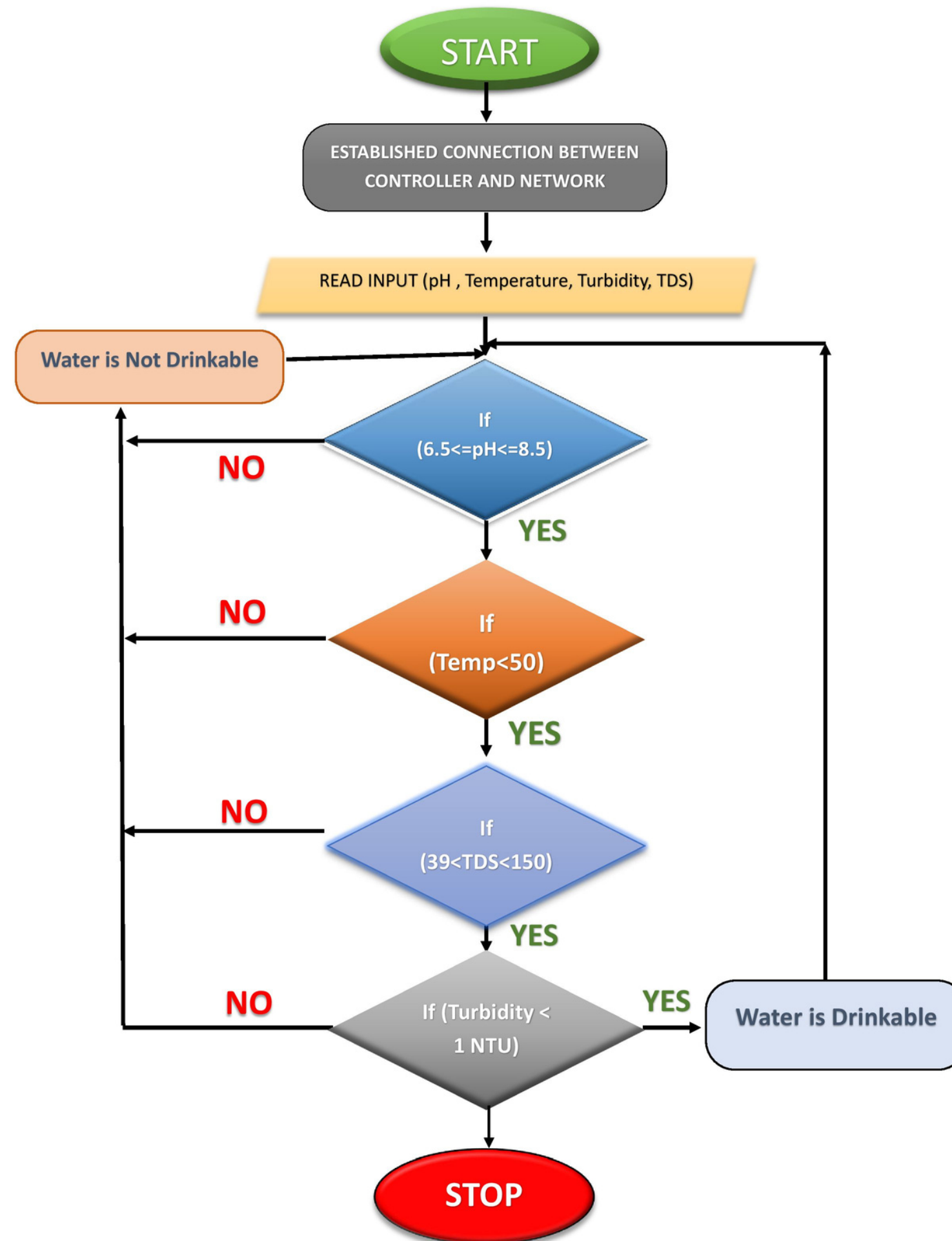
This is an 16x2 LCD display screen with I2C interface. It is able to display 16x2 characters on 2 lines, white characters on blue background. Usually, Arduino LCD display projects will run out of pin resources easily.



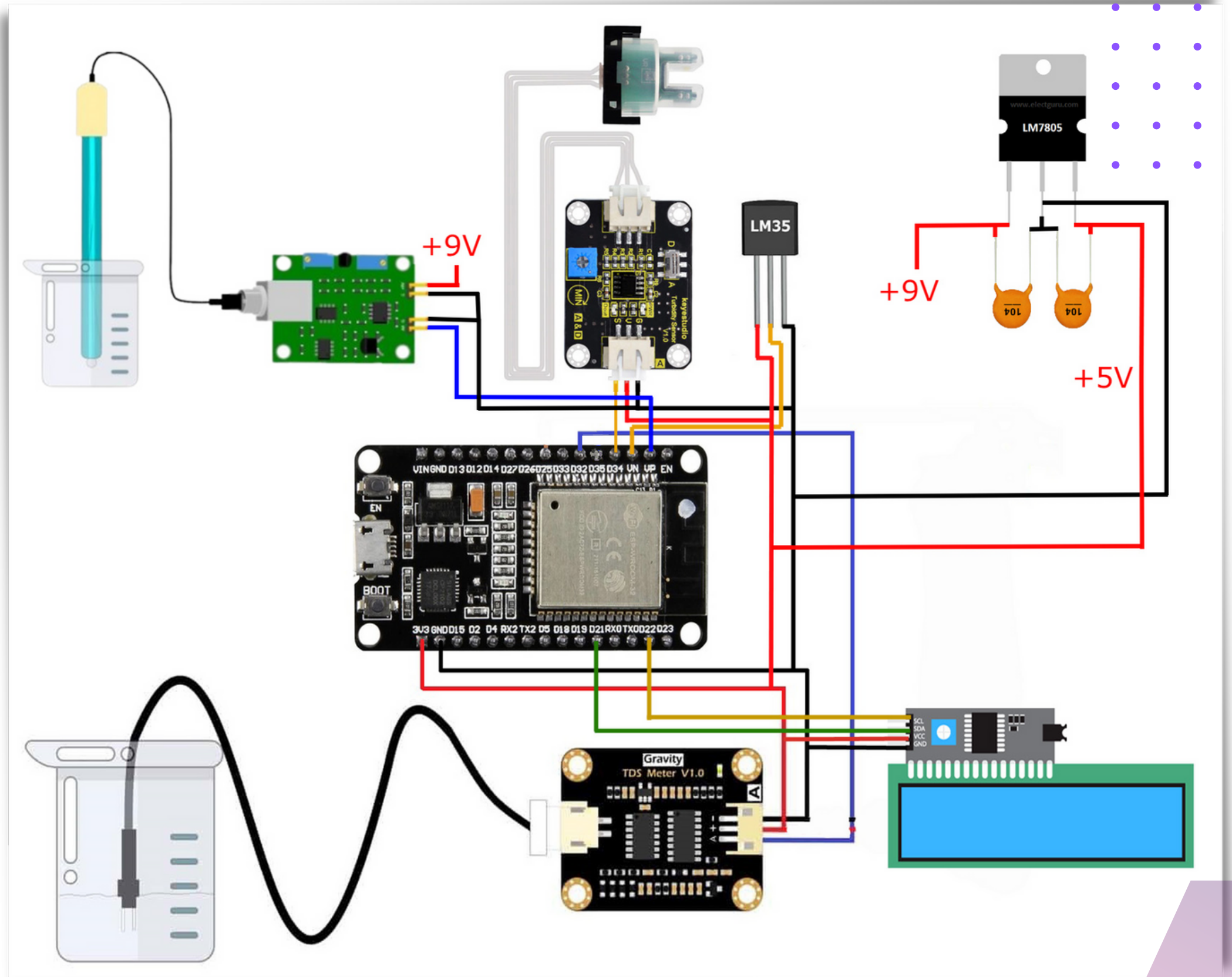
SMPS

MPS transfers power from a DC or AC source (often mains power, see AC adapter) to DC loads.
Input: 20-240VAC 50-60HZ
Output: 9V -1AMP

Flow Chart



Circuit Diagram



Simulation

Reading For Mineral Water



IoT Based Water Quality Measurement



Turbidity Value:
0.56 NTU



TDS Value: 110.21
PPM



Temperature:
28.08°C



Is Drinkable: Yes

Simulation

Reading for Mud Water



IoT Based Water Quality Measurement



Turbidity Value:
0.56 NTU



TDS Value: 0.00 PPM

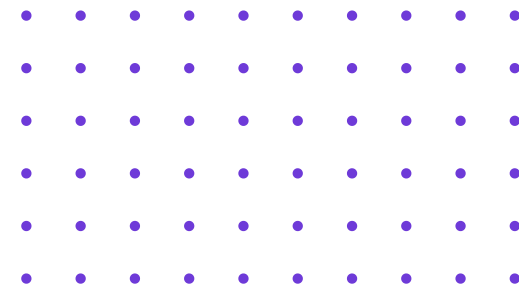


Temperature:
27.10°C



Is Drinkable: No

Application



- Pure water in healthcare facilities.
- Effective wastewater management in this day and age.
- Detecting and fixing wasteful leaks easy.
- Important for safely carrying out agricultural activities.
- Research Facilities and Laboratories.
- Wastewater Treatment.
- Aquaculture.



Advantage

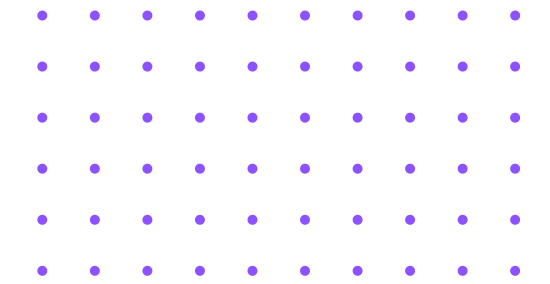


- This project is more suitable to monitor water quality parameters in real time.
- No need to visit the location to monitor the water quality parameters.
- Due to real time monitoring the water quality parameters are available whenever required which can indicate any water contamination occurred.

Disadvantage

- Accuracy of the measured value depends on the sensor used.
- Required internet connection for real time water quality parameters monitoring .

Future Scope



01

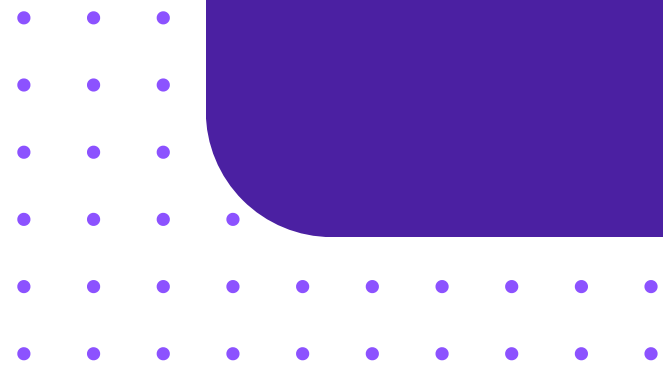
In future we increase the parameters by addition of multiple sensors.

02

Detecting the more parameters for more secure purpose.

03

By interfacing relay we controls the supply of water.



Conclusion

The system gives 80-85% accurate value.

The implementation enables sensor to provide online data to consumers. This can be improved by incorporating algorithms for detection in water quality.



Thank you!